FULMINATING MENINGOCOCCIC INFECTIONS AND THE SO-CALLED WATERHOUSE-FRIDERICHSEN SYNDROME *

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Whenever the incidence of *Neisseria meningitidis* infections approaches epidemic proportions there is an increase in the fulminating fatal infections, frequently with few or no signs of meningeal involvement. The progress of the disease in these cases is so rapid that death commonly occurs within 12 to 24 hours after the appearance of the first symptom. The presenting clinical picture is frequently one of pharyngitis, fever, and sometimes gastrointestinal symptoms, followed by the rapid development of widespread petechiae, cyanosis, peripheral vascular collapse, and death. This condition has come to be known as the Waterhouse-Friderichsen syndrome, with collapse and sudden death supposedly produced as the result of massive bilateral hemorrhage into the adrenal glands. This syndrome occasionally has been reported as occurring in other types of fulminating bacteriemia, but most frequently is associated with *N. meningitidis* infections.

In the past few years we have had the opportunity of studying 16 cases of acute fulminating meningococcic infection which were autopsied. *N. meningitidis* was recovered from cultures of either blood or cerebrospinal fluid, or both, in each case. The pathologic changes found at autopsy in these cases support the view that the presenting clinical syndrome is the result of an overwhelming bacteriemia and toxemia. It is an accepted fact that massive bilateral adrenal hemorrhage occurs in some cases, but from this series it would appear that these are only associated lesions and are not responsible in themselves for the clinical picture presented. Cases having the same clinical syndrome show no great destruction of adrenal cortical substance.

REPORT OF CASES

Case 1

M. D., a 3-months-old female, entered the hospital because of fever and rapid respiration. She had been apparently well until 12 hours before admission, when she became irritable and rapid breathing was noticed. Fever was noticed about 6 hours later. The past history and family history were negative. Physical examination before admission revealed an acutely ill infant. The temperature was 102° F.; the

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pulse rate, 170; and the respiratory rate, 100. There was no purpura. The right tympanic membrane and the throat were slightly injected. There was slight nuchal rigidity. There was dullness over the right chest with questionably impaired breath sounds and occasional fine râles over both scapular regions. The reflexes were physiologic. When seen in the hospital the infant was in extremis. Over the trunk and extremities was a fine, hemorrhagic, petechial rash. The temperature was 97° F. The child expired ½ hour later.

Summary of Gross Autopsy Findings. Purpuric hemorrhages in skin and serosae; bilateral adrenal hemorrhage; toxic changes in skin and serosae; toxic changes in spleen and kidneys; meninges negative; cultures from blood, cerebrospinal fluid, lung, pericardium, and adrenals were positive for *N. meningitidis.*

Case 2

S. L., a female, 8 years old, was admitted to the hospital because of nausea, diarrhea, and abdominal cramps of 2 days' duration, and delirium and purpura of 12 hours' duration. She had been well previously. During the preceding 2 weeks other children in the family had had "intestinal flu." Two days before admission this patient developed the same symptoms. During the evening, about 12 hours before admission, she appeared feverish and delirious. The following morning she was found to have a purpuric rash over the body and complained of indefinite joint pain and pain in the areas of the rash. On admission she appeared acutely ill, and was irrational and restless. The temperature was 101.6° F.; pulse rate, 122; respiratory rate, 40. Examination revealed mottled purplish areas of subcutaneous hemorrhage over the body, most marked on the legs. These were irregular in shape and size. There were many subconjunctival hemorrhages. There was slight pharyngeal injection and submucosal hemorrhage. There was no nuchal rigidity. The heart and lungs were normal. There was generalized abdominal tenderness and bilateral costovertebral tenderness. Reflexes were physiologic. Examination of the blood showed 15,300 white blood cells with 85 per cent polymorphonuclear cells, of which 63 per cent were nonfilamented. The erythrocytes were normal. Bleeding and clotting times were normal. Six hours after admission she appeared to be much worse. She did not respond to stimuli and the pupils were dilated. Blood pressure could not be obtained; the pulse rate was 220; and temperature, 107.2° F. A lumbar puncture was done and slightly cloudy fluid obtained. Death occurred 12 hours after admission and 24 hours after the onset of acute symptoms.

Summary of Gross Autopsy Findings. Purpuric hemorrhages in skin and mucosae; bilateral otitis media; early acute meningitis. Cultures from cerebral cortex, pericardial and peritoneal fluids, purpuric areas, and both middle ears were positive for *N. meningitidis. Diplococcus pneumoniae,* type 4, was recovered from the left middle ear.

Case 3

R. B., a male, 20 years of age, was first seen about 4 p.m., complaining of sore throat. Examination was negative except for evidence of a marked pharyngitis. The patient was seen in the hospital at about 9 p.m. Temperature at this time was 102.5° F. and there were noted a very red throat and a few scattered petechiae in the skin. There were no neurologic signs but, because of the petechiae, a blood culture was taken and a lumbar puncture done. The cerebrospinal fluid showed 130 cells, mostly polymorphonuclear leukocytes. Four grams of sulfadiazine was given

by mouth. The patient rapidly became disoriented and the petechiae increased. About 2 a.m., 10 hours after he was first seen, the patient developed respiratory failure and died.

Summary of Gross Autopsy Findings. Purpura in skin and mucous membranes; bilateral massive hemorrhage in adrenals; acute tracheobronchitis; very early meningitis. Cultures of blood and cerebrospinal fluid were positive for *N. meningitidis.*

Case 4

On the day of onset, A. B., a female, 30 years old, had felt well until afternoon when she complained of a slight sore throat. During the evening she had noticed numbness and pain in the arms and legs. There was no headache. She was seen by a physician about 10:30 p.m., when her temperature was 100.4° F.; pulse rate, 96; respiratory rate, 20. Physical examination revealed a few petechiae in the skin. There were no positive neurologic findings. The petechiae increased rapidly. The patient became drowsy and expired about 4 a.m., approximately 12 hours from the onset.

Summary of Gross Autopsy Findings. Purpura in skin and mucous membranes; bilateral hemorrhages in adrenals; toxic changes in spleen and kidneys. Cultures from blood and cerebrospinal fluid were positive for *N. meningitidis.*

Case 5

The patient, J. M., was a 1-year-old female, who was admitted to the hospital because of fever and dyspnea of 12 hours' duration. The past history was negative except for frequent colds. One week before admission she had had a "cold" with coryza and occasional cough. Twelve hours before admission she felt feverish and the cough increased in severity, causing her to vomit four or five times. Dyspnea was noted and she became drowsy. On admission she was acutely ill; respirations were rapid and grunting; there was slight cyanosis. Temperature was 106° F.; respiratory rate, 60. There were fine pinhead-sized petechiae over the neck and on one arm. Physical examination was otherwise not significant. There was no nuchal rigidity; reflexes were physiologic. The child was placed in an oxygen tent. The petechiae rapidly increased, and death occurred 1 hour after admission.

Summary of Gross Autopsy Findings. Purpura of skin and mucous membranes; bilateral hemorrhage in adrenals; acute left otitis media; toxic changes in spleen and kidneys; early meningitis. Blood culture was positive for *N. meningitidis.*

Case 6

J. F., a 22-months-old male, had been seen in a neighboring town by a physician who thought that he had bronchopneumonia. There were numerous purpuric spots over the body and extremities. The mother stated that there had been a few such areas for 2 weeks, but that they had markedly increased in the last few hours. The baby was sent to the hospital but was dead on admission.

Summary of Gross Autopsy Findings. Purpura in skin and mucous membranes; right otitis media; toxic changes in spleen, liver, and kidneys; slight hemorrhage in the adrenals; early meningitis. Cultures of cerebrospinal fluid were positive for *N. meningitidis.*

Case 7

R. W., a male, 33 years of age, was a known chronic alcoholic, employed in a restaurant. He was reported to have been intoxicated daily until the day before death. He was seen sober on that afternoon at about 3 p.m. He said that he was going to bed but did not state any reason. He left a call for 5 a.m., his usual rising hour. He did not answer the call, and later was found dead.

Summary of Gross Autopsy Findings. Purpura in skin and mucous membranes; bilateral hemorrhage in adrenals; toxic changes in liver and kidneys; culture of cerebrospinal fluid yielded *N. meningitidis;* blood culture was contaminated.

Case 8

W. T., a male, 44 years old, was admitted to the hospital in coma. There was an indefinite history of an upper respiratory infection. Otherwise he had been well until the morning of admission. At that time he became sulky and irritable, and later in the day, drowsy and incontinent of urine and feces. He was thought to be intoxicated. At 6 p.m., he was seen by a physician who noted petechiae on the body and nuchal rigidity. He was admitted to the hospital at 9 p.m. His temperature was 103° F.; pulse, 104; respiratory rate, 30; blood pressure, 80/40 mm. Hg. There were generalized petechiae and rigidity of the neck. Physical examination was otherwise negative. Lumbar puncture revealed 3840 cells per cmm. There were many Gram-negative, intracellular diplococci. Cultures of the blood and cerebrospinal fluid revealed *N. meningitidis.* Intravenous sodium sulfadiazine was given, the patient receiving 7.5 gm. in 6½ hours. He became very restless, the pulse was rapid and irregular, and death occurred 8 hours after admission and approximately 22 hours from the onset of the illness.

Summary of Gross Autopsy Findings. Purpura of skin and mucous membranes; moderate meningitis; toxic changes in spleen, liver, and kidneys.

Case 9

D. K. O., a male infant, 1½ years old, was seen by a physician because of fever and petechiae of a few hours' duration. The condition was not recognized and no specific treatment was given. The child was dead when seen several hours later.

Summary of Gross Autopsy Findings. Petechiae in skin, conjunctivae, and serous membranes; early meningitis; no gross change in adrenals. Cultures of cerebrospinal fluid were positive for *N. meningitidis.*

Case 10

The patient, E. O., was a female, 26 years of age, who had had symptoms of a mild cold for 3 days. On the afternoon of admission to the hospital there had developed fever, vomiting, chills, and stupor, followed by convulsions. On admission the patient was stuporous. Petechiae had developed over the entire body. The temperature was 102° F.; pulse, 100; respiratory rate, 46; blood pressure, 140/60 mm. Hg. There was nuchal rigidity and a bilateral Babinski sign. The cerebrospinal fluid was slightly cloudy and contained pus cells and Gram-negative intracellular diplococci resembling *N. meningitidis.* Examination of the blood showed 8,400 white cells, with polymorphonuclear cells, 80 per cent; nonprotein nitrogen, 50 mg. per cent. Blood culture was later reported as positive for *N. meningitidis.* Sodium sulfadiazine was given intravenously and a blood level of

14 mg. per cent established. The patient was given also antimeningococcic serum, adrenal cortical extract intravenously, and parenteral fluid because of low urinary output. Twenty-four hours after admission she became irrational and expired.

Summary of Gross Autopsy Findings. Petechiae of skin and mucous membranes; toxic changes in kidneys; a few small hemorrhages in the adrenals; cirrhosis of the liver. Post-mortem cultures of blood and cerebrospinal fluid were negative.

Case 11

D. K., a boy, 4½ years old, had been well until the evening before admission when he complained of headache, followed by vomiting. In the morning he was irrational and a rash was noted over his body. He was seen by a physician who found, in addition, fever and slight stupor. There was no nuchal rigidity or changes in reflexes. Later he had a convulsion, and was sent to the hospital. On entrance the temperature was 96° F.; pulse, 88; respiratory rate, 28. There was a generalized purpuric macular rash, slight cyanosis, painful rigidity of the neck, hyperactive reflexes, and positive Kernig's sign. Blood pressure was 100/40 mm. Hg. The blood examination showed the hemoglobin to be 14 gm.; red blood cells, 4.64 millions; white blood cells, 23,700; polymorphonuclear cells, 84 per cent. Urine examination revealed no significant change. The cerebrospinal fluid was cloudy, and there were many polymorphonuclear cells and many Gram-negative diplococci resembling meningococci. Cultures of blood and cerebrospinal fluid were later reported as positive for *N. meningitidis*. The patient was given sulfadiazine, adrenal cortical extract, and fluids. That evening, although he appeared somewhat improved, his temperature rose progressively to 105.2° F. and his pulse to 170 and cardiac irregularity developed. The cyanosis increased. The patient was found to be sensitive to serum and desensitization was begun. Some edema of the eyelids developed. This was partially controlled with adrenalin, but as the reaction continued, administration of antimeningococcic serum was stopped. The course was progressively worse, and the patient expired the next morning, 36 hours after admission and 60 hours after the onset of his illness.

Summary of Gross Autopsy Findings. Edema of the eyelids; petechiae in skin and epicardium; meningitis.

Case 12

M. A. was a female, 2½ years old, who was admitted to the hospital with the history of stiff neck, vomiting, and malaise of a few hours' duration. She had had a slight rhinitis for 3 days. The child was acutely ill, with rapid shallow respirations, tachycardia, and a temperature of 106.5° F. There was nuchal rigidity and a positive Kernig's sign. Examination was otherwise negative. Lumbar puncture revealed a clear fluid with no cells. Sugar and chlorides were within normal limits. Cyanosis and dyspnea increased, the temperature rose to 108° F., and the patient died 24 hours after the onset of the acute illness. There was no purpura at any time.

Summary of Gross Autopsy Findings. Pulmonary congestion; congestion of brain; adrenals, negative; blood culture, positive for *N. meningitidis*.

Case 13

E. B., a male, 34 years of age, was admitted to the hospital because of pain in the legs, dyspnea, vomiting, diarrhea of 19 hours' duration, and a purpuric rash of 9 hours' duration. Past history was negative. On examination his temperature

was 100.6° F.; pulse, 100; respiratory rate, 28. There was restlessness and severe dyspnea. A diffuse, blotchy, purpuric rash was present over most of the body. There was no nuchal rigidity. The blood pressure could not be obtained. Neurologic examination was negative. The patient was given adrenal cortical extract intravenously and sulfamerazine orally. The blood pressure was recorded as 90/70 mm. Hg, but fell progressively to 58/45 and later could not be read. The temperature rose progressively to 104.6° F. The patient became very cyanotic and expired 12 hours after admission and 31 hours after the onset of the acute disease. An ante-mortem blood culture was reported as positive for *N. meningitidis*.

Summary of Gross Autopsy Findings. Purpuric rash over entire body; petechiae in conjunctivae and serous membranes; massive hemorrhage in adrenals; cloudy cerebrospinal fluid.

Case 14

A. M. was a male, 53 years old, who had complained of weakness of the lower extremities and nausea 2 hours before admission. His family found him on the floor, unable to talk. A convulsion followed and he was sent to the hospital. On admission his temperature was 99.6° F.; pulse, 84; respiratory rate, 26; blood pressure, 85/65 mm. Hg. The only positive physical findings were slight spasticity and hyperactive reflexes in the upper extremities. The peripheral blood was not remarkable. The urine showed albumin, 2 plus, with a few granular casts and pus cells microscopically. Cerebrospinal fluid pressure was 210 mm. of water. The fluid was cloudy and contained 850 cells per cmm. No organisms were seen in smears. Twenty-one hours after admission petechiae in the skin were noted. The temperature was 102° F. The blood pressure remained around 95/80. The patient became progressively worse and died 28 hours after admission.

Summary of Gross Autopsy Findings. Petechiae on the torso and lower extremities, few in serous membranes; lungs, congested; spleen, soft; no gross hemorrhage in adrenals; meningitis, 2 plus; cerebrospinal fluid positive for *N. meningitidis* by culture.

Case 15

R. M., a 1-year-old male infant, had had a mild upper respiratory infection for 2 days. Approximately 24 hours before admission the child was noted to be restless. This was followed in a few hours by vomiting and a red rash on the skin. The child was sent to the hospital. The temperature was 103° F.; pulse, 165, weak and rapid; respiratory rate, 65; blood pressure, 145/100 mm. Hg. The pharynx was red. There was nuchal rigidity, hyperactive reflexes, and bilaterally positive Kernig's sign. Lumbar puncture showed increased spinal fluid pressure, and a cell count of 3600 per cmm., 60 per cent polymorphonuclear leukocytes and 40 per cent lymphocytes. Chlorides, 680 mg. per 100 cc. Meningococci were found in the fluid. The peripheral blood was not remarkable. The child was treated with penicillin intravenously and intrathecally and given sulfamerazine by mouth. Six hours after admission he was lethargic and semistuporous. Temperature was 104° F.; pulse, 175; respiratory rate, 85; blood pressure, 90/60. There were increasing numbers of petechiae, and extreme cyanosis. Death occurred 8 hours after admission.

Summary of Gross Autopsy Findings. Petechiae over upper body, few in the serous membranes; blood-stained fluid in left pleural cavity;

congestion and edema of lungs; congestion of kidneys; negative adrenals; meningitis.

Case 16

I. L., a boy, 5 years old, was seen by a physician because of nausea and vomiting, fever, and a skin rash of 12 hours' duration. He had had a slight "cold" for several days. On admission to the hospital, the temperature was 103.6° F.; pulse, 200; respiratory rate, 20. The blood pressure could not be obtained. The child was deeply cyanotic and in coma. There were numerous petechiae in the conjunctivae and over the entire body, and erythema of the thighs. There was no nuchal rigidity. The reflexes were absent. Death occurred 1 hour after admission, approximately 13 hours after the acute onset.

TABLE I

Pertinent Clinical Data in 16 Cases of Acute Fulminating Meningitis

Case	Sex	Age	First symptoms	First temp.	Purpura	Blood culture	Spinal fluid culture	Duration to death
				°F.				*hours*
1	F	3 mos.	Fever; irritability	102°	+++	+	+	12
2	F	8 yrs.	Nausea; diarrhea	101.6°	+++	+	+	24
3	M	20 yrs.	Pharyngitis	102.5°	++++	+	+	10
4	F	30 yrs.	Pharyngitis	100.4°	++++	+	+	12
5	F	1 yr.	Fever; dyspnea	106°	++	+	o	13
6	M	22 mos.	Fever	—	+++	o	+	Few
7	M	33 yrs.	?	—	+++	o	+	Few
8	M	44 yrs.	Pharyngitis	103°	++++	+	+	22
9	M	1½ yrs.	Fever	—	++++	o	+	Few
10	F	26 yrs.	Fever	105°	+++	+	+	24
11	M	4½ yrs.	Headache; vomiting	—	++	+	+	60
12	F	2½ yrs.	Malaise; stiff neck	106.5°	o	+	o	24
13	M	34 yrs.	Dyspnea; vomiting	100.6°	++++	+	o	31
14	M	53 yrs.	Nausea; weakness	99.6°	++	o	+	30
15	M	1 yr.	Vomiting; restlessness	103°	+++	o	+	32
16	M	5 yrs.	Vomiting; fever	103.6°	++++	+	+	13

Summary of Gross Autopsy Findings. Petechiae in the skin, numerous petechiae in peritoneum and intestinal tract; bilateral hemorrhagic adrenals; early meningitis; petechiae in brain and meninges.

A summary of the pertinent clinical data is given in Table I.

PATHOLOGIC FINDINGS

A summary of the microscopic changes is found in Tables II and III.

TABLE II

Gross and Microscopic Changes in 16 Cases of Acute Fulminating Meningitis

Case	Brain	Choroid plexus	Heart	Lungs	Spleen	Liver
1	Negative		Some Zenker's degeneration	Congestion; slight hemorrhage; slight thrombosis	Congestion	Cloudy swelling
2	Meningitis, 2+	Congestion	Marked Zenker's degeneration; acute reaction; thrombosis	Congestion; hemorrhage; rare thrombus; bronchitis	Congestion	Congestion
3	Meningitis, 1+	Focal acute reaction; marked thrombosis	Marked patchy acute necrosis with polymorphonuclear reaction	Congestion; hemorrhage; few thrombi; acute bronchitis	Splenic tumor	Congestion; swollen endothelium
4	Negative		Focal acute necrosis with polymorphonuclear reaction; capillary thrombi	Congestion; capillary thrombi	Congestion	Few fine hyaline thrombi
5	Meningitis, 1+		Few small inflammatory foci	Congestion; few capillary thrombi, one in small arteriole	Slight necrosis in germinal centers	Many leukocytes in sinuses
6	Meningitis, 1+		Slight Zenker's degeneration; slight polymorphonuclear reaction	Congestion; hemorrhage; acute bronchitis	Congestion	Negative
7	Negative		Few foci of polymorphonuclear and mononuclear cells	Extensive thrombosis of capillaries; small infarcts; hemorrhage, 4+	Slight splenic tumor	Few necrotic liver cells

8	Meningitis, 2+		Moderate foci of polymorphonuclear and mononuclear cells	Extensive thrombosis of capillaries; acute bronchitis	Slight splenic tumor	Congestion; few necrotic cells
9	Meningitis, 1+	Acute reaction; thrombosis	Slight Zenker's degeneration	Congestion	Slight splenic tumor	Fatty change
10	Negative	Early capillary thrombosis	Slight Zenker's degeneration; slight polymorphonuclear reaction	Congestion; few capillary thrombi	Slight splenic tumor	Cirrhosis; fatty change
11	Meningitis, 4+		Slight Zenker's degeneration	Slight bronchopneumonia	Congestion	Fatty change
12	Meningitis, 1+; thrombosis		Slight polymorphonuclear and mononuclear reaction; one thrombus	Congestion and edema	Necrosis in germinal centers	Area of sinus thrombosis
13	Cloudy spinal fluid		Foci of necrosis and polymorphonuclear reaction	Few capillary thrombi	Slight splenic tumor	Central necrosis
14	Meningitis, 3+	Acute reaction	Marked polymorphonuclear and mononuclear reaction	Marked thrombosis; acute bronchitis	Acute splenic tumor	Thrombosis in sinusoids
15	Meningitis, 3+	Thrombosis and acute reaction	Slight Zenker's degeneration; one thrombus	Congestion	Marked necrosis of germinal centers	Fatty change
16	Meningitis, 1+; few thrombi		Inflammatory foci	Congestion and edema	Necrosis of germinal centers	

TABLE III

Renal and Adrenal Lesions, Thrombosis, and Gross Hemorrhage in the Adrenals, Tabulated for 16 Cases of Acute Fulminating Meningitis

Case	Kidneys	Adrenals	Summary of thrombosis	Gross adrenal hemorrhage
1	Extensive hyaline thrombosis; capsular hemorrhage	Marked hemorrhage; slight acute reaction in medulla; few thrombi	Lungs, kidneys, adrenals	4+
2	Hyaline degeneration of basement membrane; one capillary thrombus	Acute reaction in medulla; thrombi in border capillaries; very slight hemorrhage	Heart, lungs, kidneys, adrenals	0
3	Marked thrombosis; early tubular degeneration	Marked hemorrhage and necrosis; extensive hyaline thrombosis	Lungs, kidneys, adrenals, pancreas, skin	4+
4	Moderate thrombosis; hyaline tubular degeneration	Necrosis and polymorphonuclear infiltration of medulla; areas of cortical necrosis; congestion and hemorrhage	Heart, lungs, liver, kidneys	4+
5	Few areas of hyaline thickening of basement membrane; few thrombi	Congestion and hemorrhage; polymorphonuclear infiltration of medulla; thrombi in large vessels and cortical capillaries	Adrenals; slight in lungs and kidneys	2+
6	Few questionable thrombi in tufts	Intense acute reaction in medulla; slight hemorrhage; moderate thrombosis	Adrenals; questionable in kidneys	1+

7	Marked capillary thrombosis; early tubular degeneration	Extensive hemorrhage and necrosis; focal acute reaction in surviving areas	Marked in lungs and kidneys	4+
8	Early hyaline thrombi in tufts; débris in tubules	Focal mononuclear and polymorphonuclear cells in medulla; thrombus in one vein; slight hemorrhage	Lungs, kidneys, adrenals	o
9	Slight hyaline degeneration in tufts; fatty degeneration in convoluted tubules	Marked capillary thrombosis; congestion; slight hemorrhage	Choroid plexus, adrenals	o
10	Hyaline degeneration; moderate thrombosis	Slight polymorphonuclear reaction in medulla	Choroid plexus, lungs, kidneys	1+
11	Degeneration of tuft endothelium	Negative	None	o
12	Slight hyaline change in tufts	Necrosis and polymorphonuclear reaction in medulla; cortical necrosis; few thrombi	Brain, heart, liver, adrenals	o
13	Moderate capillary thrombosis; tubular degeneration	Hemorrhage; thrombosis; focal necrosis and polymorphonuclear reaction	Lungs, kidneys, adrenals	4+
14	Congestion	Loose fibrin thrombi in central veins	Lungs, liver, adrenals	o
15	Congestion	Necrosis and polymorphonuclear reaction in medulla; thrombosis and slight hemorrhage	Choroid plexus, heart, adrenals	o
16	Congestion	Marked congestion and hemorrhage	Brain	4+

Skin. Hemorrhage into the skin varied from moderate to marked in all cases except one, in which there was none (Fig. 1). Sections through areas of petechiae were available in only 2 cases. In one, diffuse hemorrhage was seen in the corium, without recognizable changes in the blood vessels. In the other, rather diffuse hemorrhage and one area of capillary thrombosis were seen.

Heart. The hearts were not unusual in size. In 12 of the 16 cases, petechial hemorrhages were found in the epicardium and beneath the endocardium. In the epicardium these tended to involve the distribution of the coronary arteries, but scattered hemorrhages were present, also. The endocardial hemorrhages were irregular in distribution. In 6 cases the myocardium was more flabby than usual. Microscopically, all but 4 cases showed varying degrees of leukocytic reaction, either perivascular or in the areas of degeneration, which was largely polymorphonuclear; but in some instances a mixture of polymorphonuclear neutrophils, eosinophils, and mononuclear cells was present (Figs. 2 and 3). In 5 there were definite areas of necrobiotic change of heart muscle, varying from pyknosis of nuclei to complete necrosis of a number of cardiac muscle cells. In 4 cases thrombi were found in capillaries in the myocardium.

Lungs. Grossly, the lungs were not unusual aside from showing varying degrees of congestion and edema. This was evident also microscopically. Actual hemorrhage into interstitial tissue and alveoli varied from slight to marked in all cases. In 5 there was histologic evidence of acute bronchitis, in one instance the exudate was hemorrhagic. Only 2 cases showed evidence of exudation into the alveoli. Hyaline thrombi in capillaries of the lung were commonly seen in 11 of the 16 cases. In 3 they were noted as being slight, in 5 as being moderate, and in 3 as being extensive in degree. In one instance a hyaline thrombus was found in one of the small arterioles.

Spleen. Changes in the spleen were much more marked in the adults than in the children. All of the children's spleens were of normal size, although one appeared slightly softer than usual. In the adults, 6 of the 7 cases showed the gross and microscopic changes associated with early acute splenic tumor and one appeared normal. In some of the children the germinal centers of the splenic corpuscles appeared hyperplastic, but this was not constant. In 4 there was evidence of necrosis of the germinal centers.

Liver. Gross changes in the liver were not remarkable. Microscopically, no uniform change was found. In one case there was rather extensive central necrosis with mononuclear cell reaction. In several there was a slight degree of fatty metamorphosis of the liver cells.

However, in 3 cases there were found areas of hyaline thrombosis in the sinusoids.

Pancreas. Gross and microscopic examination of the pancreas was negative except for the presence of a few hyaline capillary thrombi in 2 cases; one of these was in a child, the other in an adult.

Kidneys. Gross changes in the kidneys were slight but rather uniform. In all cases there appeared slight pouting of the cut surfaces of the kidney substance. The markings tended to be indistinct. In some the kidney substance had a slightly cooked appearance. Punctate areas of hemorrhage into the mucous membranes of the pelves were common. Microscopically, changes in the kidney tissue were rather marked. The predominant lesion was one of hyaline thrombosis in the capillaries of the glomerular tufts. This was present to some degree in 10 of the 16 cases (Fig. 5). In many sections most of the capillary loops of certain glomeruli were occluded by massive reddish hyaline thrombi. In other instances such hyaline material seemed to be present along the capillary walls without completely occluding the vessel. Where thrombosis was present to a less marked degree it was possible to see the fibrinous nature of the material (Fig. 6). In addition to the thrombi in many of the glomeruli, there were scattered areas of hyaline degeneration, apparently involving the endothelial cells of the capillary loops, without thrombosis being present. Exudation of leukocytes into these involved capillary tufts was not prominent, but occasionally polymorphonuclear cells were seen. The tubules were affected in varying degrees. In some there was simple swelling of the cells of the convoluted tubules; in 2 there was rather extensive fatty degeneration of tubular epithelium; in 5 of these cases the tubules contained relatively large quantities of desquamated epithelium and débris.

Adrenals. Gross changes were present in the adrenals in 9 of the 16 cases. In all 9, edema of the perirenal fat and capsule was noted. The adrenals in the involved cases were moderately increased in size but in general maintained their normal outline. On section, reddish purple discoloration of varying degree was found scattered throughout the gland, in some cases almost completely replacing the normal markings. Occasionally, a rim of normal-appearing cortical tissue could be seen around the periphery of the gland substance.

Microscopically, there was only one case in which no definite change could be found in sections of adrenal gland. This was in the child who survived 60 hours. In all other cases, even though the glands appeared grossly normal, definite and usually rather marked microscopic changes were present. These varied in both type and degree. The simplest change consisted of focal areas of necrosis involving the medullary

portion of the gland with an infiltration of polymorphonuclear and eosinophilic leukocytes. In other adrenals this process of acute necrosis and polymorphonuclear infiltration was so marked as to replace the medullary tissue almost completely (Figs. 7 and 8). Focal areas of necrosis and polymorphonuclear infiltration were seen occasionally in the cortical tissue, but these were not prominent. Congestion and hemorrhage were present in these cases to only a slight degree. Hemorrhage was found mostly in the outer areas of the medulla and extended to the inner layers of the cortex. In cases showing more marked involvement, it was so great as to destroy almost completely all normal markings. Occasionally, in the surviving areas, foci of necrosis of both cortex and medulla could be seen. Thrombosis of the adrenals was seen in 11 cases. It was most prominent in the capillaries between the medulla and the cortex (Figs. 9 and 10). In some instances it was massive and extended along the sinusoids between the cords of cortical tissue. In 2 instances both thrombosis and hemorrhage were so marked that the picture suggested massive infarction of cortical tissue. Thrombosis of the central vein of the adrenals was seen in 2 cases (Fig. 4). These thrombi were similar to those seen in the kidney. Where they were present to a slight degree the fibrin constituent could be readily recognized. In areas where they were present to a more marked degree, they were reddish and hyaline. Many of these thrombi could be missed very easily in examination of sections prepared with the routine hematoxylin and eosin stain. They became very prominent when Masson's trichrome stain was used.

The tubular degeneration of the adrenal cortex described by Thomas,[1] Dietrich,[2] and others, and recently emphasized by Rich,[3] was seen in some cases but was not uniform. It was present to the greatest degree in those cases showing the most marked gross and microscopic changes, and did not explain the circulatory collapse in those cases showing minimal adrenal lesions (Table IV).

Sections of the *lymph nodes, thymus, gastrointestinal tract, diaphragm, gallbladder, urinary bladder, aorta,* and *thyroid* showed nothing significant except for petechial hemorrhages and an occasional capillary thrombus.

Brain. The brain was examined in 15 of the 16 cases. In 5 there was no gross evidence of meningitis. However, in one of these microscopic examination showed a slight infiltration of leukocytes in the pia arachnoid. Congestion of the pia arachnoid and of the brain substance was rather pronounced. In the 10 cases with gross evidence of exudate into the pia arachnoid, microscopic examination revealed the usual histologic changes of acute leptomeningitis. In the one case

(no. 13) in which the head was not opened, a spinal tap was performed at autopsy and slightly cloudy fluid was removed from the subarachnoid space. Focal hemorrhages into the pia arachnoid were common even in the absence of meningeal exudate. In 2 cases microscopic examination revealed several areas of hyaline capillary thrombosis in vessels in the brain tissue.

Choroid Plexus. Sections of the choroid plexus were available in 6 cases. One showed simple congestion. In 4 there was a rather diffuse but focal inflammatory reaction consisting largely of polymorphonu-

TABLE IV

Degree of Meningitis and Adrenal Changes with Available Blood Pressure Readings

Case	Degree of meningitis	Blood pressure	Degree of adrenal hemorrhage	Tubular degeneration	Survival
		mm. Hg			*hours*
1	o	—	++++	o	12
2	++	Too low to read	o	++++	24
3	+	—	++++	+	10
4	o	—	++++	++++	12
5	+	—	++	o	13
6	+	—	+	o	Few
7	o	—	++++	++	Few
8	++	80/40	o	++	22
9	+	—	o	+	Few
10	o	140/60	+	+	24
11	++++	100/40	o	+	60
12	+	—	o	o	24
13	++ (fluid)	90/70 to 58/45	++++	o	31
14	+++	85/65	o	++++	30
15	+++	145/100 to 90/60	o	+	32
16	+	Too low to read	++++	o	13

clear leukocytes, and hyaline and fibrin thrombi were present in the capillaries of the plexus (Figs. 11 and 12).

Middle Ears. Acute otitis media was found as an associated lesion in 3 of the children.

BACTERIOLOGIC PROCEDURES

It will be noted that *Neisseria meningitidis* was isolated from either the blood or cerebrospinal fluid, or both, in every case in this series. Such successful results are obtained by careful attention to modern bacteriologic detail. The centrifugated sediment from the cerebrospinal fluid is inoculated upon at least three different media; an entire "chocolate" agar plate is prepared from infusion agar base by the addition of 6 per cent citrated horse's blood and heated at 85° C. for 5 minutes, and an entire fresh blood agar plate prepared from infusion agar base by the addition of 6 per cent citrated horse's blood. The contents (5 cc.) of a dextrose semisolid fermentation tube are then

mixed with the remaining sediment, usually in the original specimen tube. This medium is Difco * phenol red broth base to which 0.5 per cent dextrose and 0.2 per cent agar have been added. The dilution of the sediment under such conditions prevents the inhibition of multiplication of the organisms by the enzymes of the leukocytes present and allows for successful isolation in certain instances in which the ordinary cultural methods fail. It is not unusual to obtain a heavy growth of this organism in such tubes in 18 to 24 hours, while a very sparse growth is obtained on the plates at 48 hours. All cultures are incubated at 35° to 36° C.

The blood cultures are prepared by adding approximately 7 to 10 cc. of blood to 70 cc. of medium contained in a rubber-capped bottle under reduced oxygen tension. The medium is a buffered tryptose phosphate broth (Difco *) containing 0.1 per cent agar, 0.1 per cent sodium citrate, and 0.02 per cent para-amino-benzoic acid. If the patient has been receiving penicillin therapy, penicillinase in adequate amounts is added at the time of collection of the blood. These cultures are incubated at 35° to 36° C. and inspected daily or twice daily for evidence of growth.

Discussion

In the 16 cases of overwhelming meningococcic infection presented, there was considerable variation in both the clinical syndrome and in the post-mortem findings.

Of these 16 cases, 10 were males and 6 were females. The age incidence varied from 3 months to 53 years. Nine of these cases were in children ranging in age from 3 months to 8 years, while 7 were adults ranging from 20 to 53 years of age. Although the picture of overwhelming meningococcemia is one which is recognized by pediatricians, the rather common occurrence of this same condition in adults has not been generally recognized. Moritz and Zamcheck [4] stated: "The incidence of rapidly fatal meningococcemia appears to be considerably higher in younger than in older soldiers." One of the most striking features of the disease is the rapidity of its progress. Although several of these patients had complained of an ordinary upper respiratory infection for a period of 2 or 3 days, most of them (or their parents) could place the onset of acute illness at some particular hour of the day or night. From this time on, 12 of the 16 cases ended fatally within 24 hours or less. Three patients survived about 30 hours, and one survived approximately 60 hours. This confirms the frequently quoted statement of Herrick [5]: "No other infection so quickly slays." Although the presenting symptoms varied considerably, it was notable

* Difco Laboratories, Detroit, Michigan.

that in adults pharyngitis was a common chief complaint, whereas in children fever, headache, and irritability appeared to be particularly common. In patients who were seen by physicians, cyanosis was marked.

Purpura of the skin and mucous membranes was prominent in all except one case. These hemorrhagic manifestations developed rapidly and could be seen to increase rapidly while the patient was under observation. Increasing cyanosis, delirium, stupor, circulatory collapse with falling blood pressure and death soon followed. Cases of this type have been known as the Waterhouse-Friderichsen syndrome. Martland [6] found only 107 cases of this syndrome prior to 1943 and added 19. Many have been described since.

In most cases the causative organism has been a meningococcus, but other organisms such as *Diplococcus pneumoniae, Streptococcus haemolyticus* (beta), *Haemophilus influenzae,* and others have been reported. In all of our cases the meningococcus was isolated from either the blood or cerebrospinal fluid, or both. This is quite in contrast with the statement of Rucks and Hobson [7] that "search for the causative agent in the spinal fluid is generally fruitless." The reports of McLean and Caffey,[8] and others, and more recently of Tompkins [9] on the recovery of organisms from the purpuric lesions offer another method for the rapid recognition of the inciting factor.

The striking change seen at autopsy, in addition to those in the meninges and adrenals, is the microscopic evidence of diffuse and marked vascular damage with thrombosis in many organs of the body. Thrombi were found in the capillary systems of various organs in all except one case—in the patient who survived for 60 hours. In the more severe areas of involvement almost every vessel, in some organs, was occluded by such masses of hyaline material. Such changes were seen in the heart, lungs, liver, kidneys, adrenals, pancreas, and gastrointestinal tract in varying degrees. Hill and Kinney [10] have recently described the marked vascular damage and thrombosis which they found in the skin, membranes, and other organs. In the present series there was no correlation between the amount or degree of thrombosis in one organ and in another, or between the occurrence of such thrombosis and the presence of meningitis. Extensive thrombosis was found in the organs of some patients in whom changes in the adrenal glands were comparatively slight. In others, massive thrombosis occurred in the kidneys and adrenals, and other organs showed comparatively slight changes. In 7 of these cases there was no gross hemorrhage into the adrenals and in 3 it was slight to moderate. Even in cases in which it was marked, much surviving adrenal cortical tissue was seen when

the glands were examined microscopically. In those cases in which the blood pressure was taken there was no correlation between the degree of adrenal cortical destruction and the level of the blood pressure (Table IV). In those cases in which it was not taken, circulatory collapse was evident. Clinically, and otherwise at autopsy, the cases with and without massive hemorrhage into the adrenals cannot be separated. Boger [11] stated: "Clinically, the term 'fulminating meningococcemia' seems preferable to 'Waterhouse-Friderichsen syndrome,' and if the latter has any usefulness it should be restricted to pathological discussions." From these studies its use does not appear justified in pathologic discussions. There is evidence of an intense overwhelming bacterial infection with marked vascular damage resulting in thrombosis, hemorrhage, or both. Such changes were found in varying degree in every organ examined, although, as would be expected, not uniformly in all organs in all cases. In some patients the damage in one organ was more marked than in another. The reason for this is unknown, but such unexplained variations in degree of involvement are not unusual in other infections. However, from the evidence presented, there appears no reason for selecting one organ over another for special attention. It is true that, clinically, the collapse that is seen in patients with fulminating meningococcemia simulates the collapse of acute adrenal cortical insufficiency, and that in some cases of such meningococcemia there is massive but not total destruction of adrenal cortical tissue. However, the occurrence of many similar cases without massive involvement of the adrenal cortex appears to minimize the importance of those changes as the productive cause of the clinical syndrome.

There appears to be no correlation between other described changes in the cortex and the degree of circulatory collapse. It is still possible that exhaustion of the cortical cells may be present without histologic changes being produced. It appears probable that such exhaustion takes place more readily in the presence of overwhelming infection, but at the present time there are no criteria for the recognition of such exhaustion. For these reasons we believe the term "Waterhouse-Friderichsen syndrome" should be discontinued. Either the term "fulminating meningococcic infection" or "fulminating meningococcemia" appears justified. In this series, blood cultures taken either antemortem or post-mortem were positive in 11 of 16 cases. One was contaminated, and one was taken only after rather vigorous treatment had been given (case 15). The term "meningococcic meningitis" does not sufficiently separate the fulminating cases from those commonly

seen, and in this series there was no evidence of meningitis at the time of death in 4 cases.

Some clinicians have attempted to divide cases of fulminating meningococcemia into adrenal and meningeal types.[12] As seen in Table IV, in this series, although occasional cases showed marked change in one location and not the other, there is no constant relationship.

Treatment

Considering the overwhelming infection and toxemia in such cases, the value of treatment is a matter of great interest. The fact that patients presenting the classical picture of overwhelming meningococcic infection and intoxication have survived following prompt recognition and vigorous therapy, stimulates one to advocate the use of all available therapeutic agents in adequate dosage. Sulfadiazine by mouth may be satisfactory in the ordinary case of meningococcic meningitis, with or without bacteriemia, but this drug should be used intravenously, in adequate dosage, if the greatest value is to be obtained in the fulminating type of this infection. It is quite apparent that penicillin should be used in conjunction with sulfonamide therapy and that in the cases showing evidence of meningeal involvement it should be administered intrathecally as well as parenterally in adequate and sustained dosage. The use of antimeningococcic serum has been practically discontinued in the treatment of meningococcic meningitis. It is apparent, however, that in the fulminating case the patient needs assistance in combating the overwhelming infection and toxemia until the defense mechanisms can produce sufficient immune substance to sensitize properly its soluble products. The use of antimeningococcic serum in adequate dosage is definitely indicated in the suspected presence of severe infection and intoxication. Adrenal cortical extract has been used by those who believe that the peripheral vascular collapse is due to adrenal damage. At the present time its use must be considered empiric. In this series, cases 2, 4, 6, 9, and 11 were not recognized clinically. Cases 3, 12, and 13 were recognized or suspected, but not adequately treated. Cases 1, 5, 7, and 16 were seen too late for any treatment to be of possible use, and only 4 patients (cases 8, 10, 14, and 15) were given what appeared to be adequate treatment.

Summary

Sixteen fatal cases of fulminating meningococcic infection were studied in which complete post-mortem and bacteriologic examinations were made.

During epidemics, meningococcic infection of this type is not uncommon, and the condition must be recognized and treated properly and early if the mortality rate is to be diminished.

The findings at autopsy are those of an overwhelming bacterial infection with vascular damage, thrombosis and hemorrhage in many organs. Meningeal involvement may be slight or absent.

The term "Waterhouse-Friderichsen syndrome" should be discontinued as evidence shows the condition to be one of general bacterial toxemia, and the occurrence of massive hemorrhage into the adrenal glands is not necessary to produce the peripheral vascular collapse which is so prominent in these cases.

Miss Anne Moran, Assistant Bacteriologist, Bureau of Laboratories, Syracuse Department of Health, and Miss Winifred Osborne, Research Assistant, Department of Bacteriology and Parasitology, rendered valuable technical assistance in the bacteriologic studies herein reported. Grateful acknowledgment is due Miss Stella Zimmer for the photomicrographs.

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[*Illustrations follow*]

DESCRIPTION OF PLATES

PLATE 123

FIG. 1. Cutaneous purpura. Lesions vary from punctate areas as on the hand to more massive areas as seen on the legs. Present in all cases except one (no. 12).

FIG. 2. Myocardium, showing a lesion of most extensive form. Arteritis with thrombosis and marked perivascular infiltration of neutrophils, eosinophils, and large mononuclear leukocytes. Hematoxylin and eosin stain. × 240.

1



2

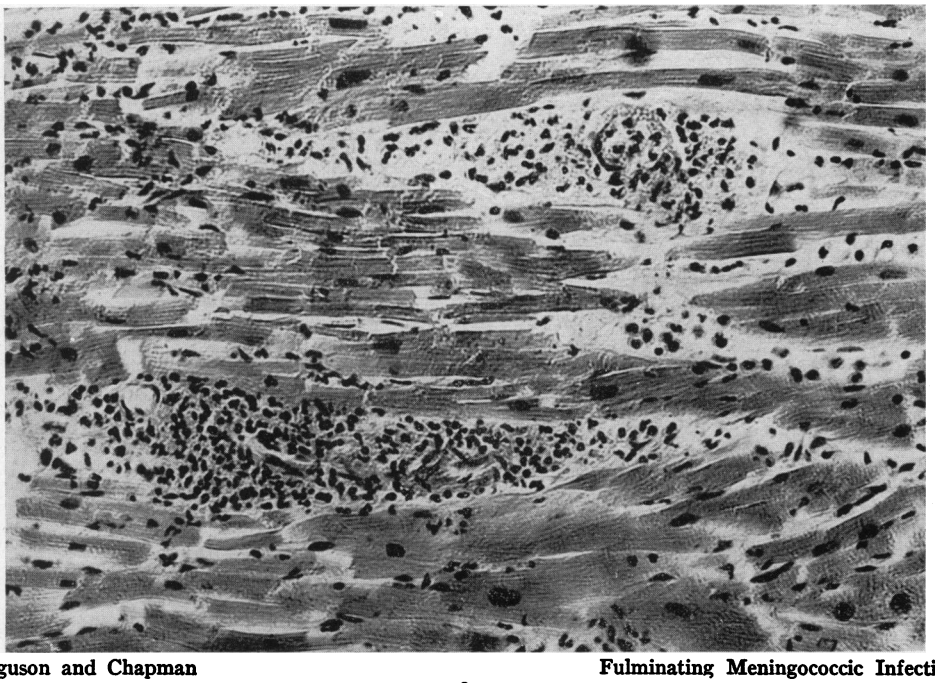
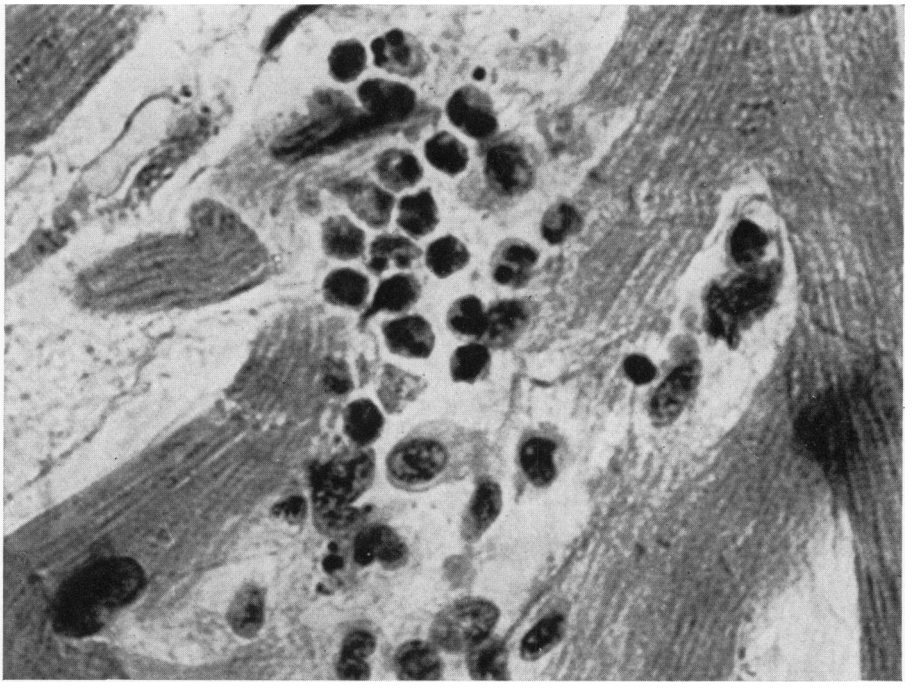


PLATE 124

FIG. 3. Myocardium. Focal area of inflammatory infiltration. Neutrophils, eosinophils, and large mononuclear leukocytes, many with eosinophilic cytoplasm. Myocytes were seen in many fields. Hematoxylin and eosin stain. × 950.

FIG. 4. Adrenal medulla. Mixed thrombus in central vein. Goldner's modification of Masson's trichrome stain. × 230.

3



4

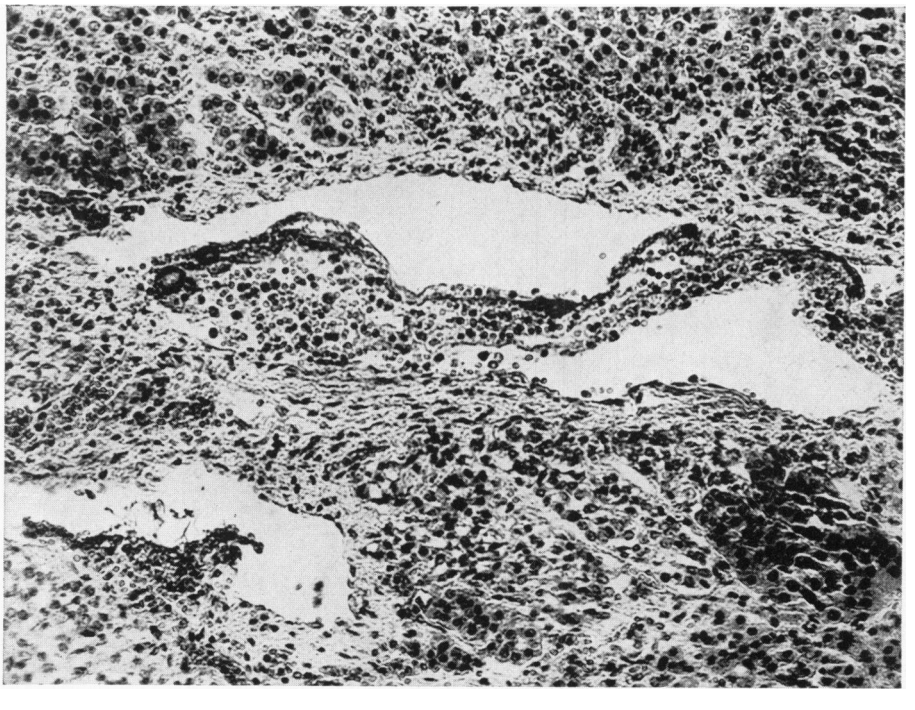
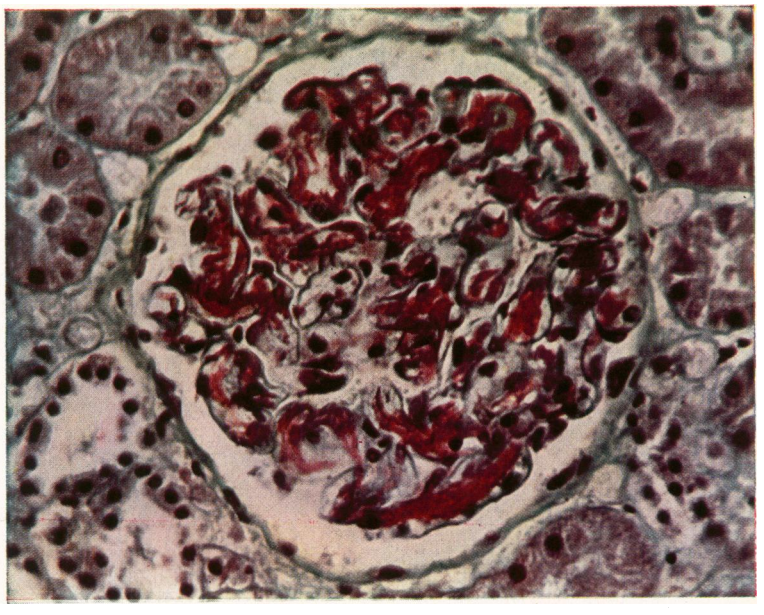


PLATE 125

FIG. 5. Kidney. Marked hyaline thrombosis in glomerular tuft. In some areas the capillary is completely occluded. In other areas the material is attached to the endothelium, leaving a central space. Goldner's modification of Masson's trichrome stain. × 420.

FIG. 6. Higher power of the glomerular tuft seen in Figure 5, showing location and character of hyaline material. Goldner's modification of Masson's trichrome stain. × 970.

5



6

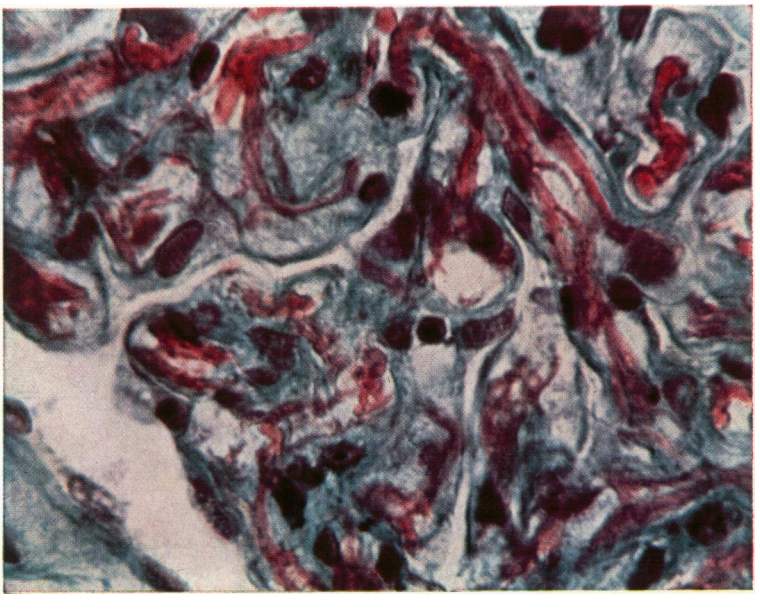
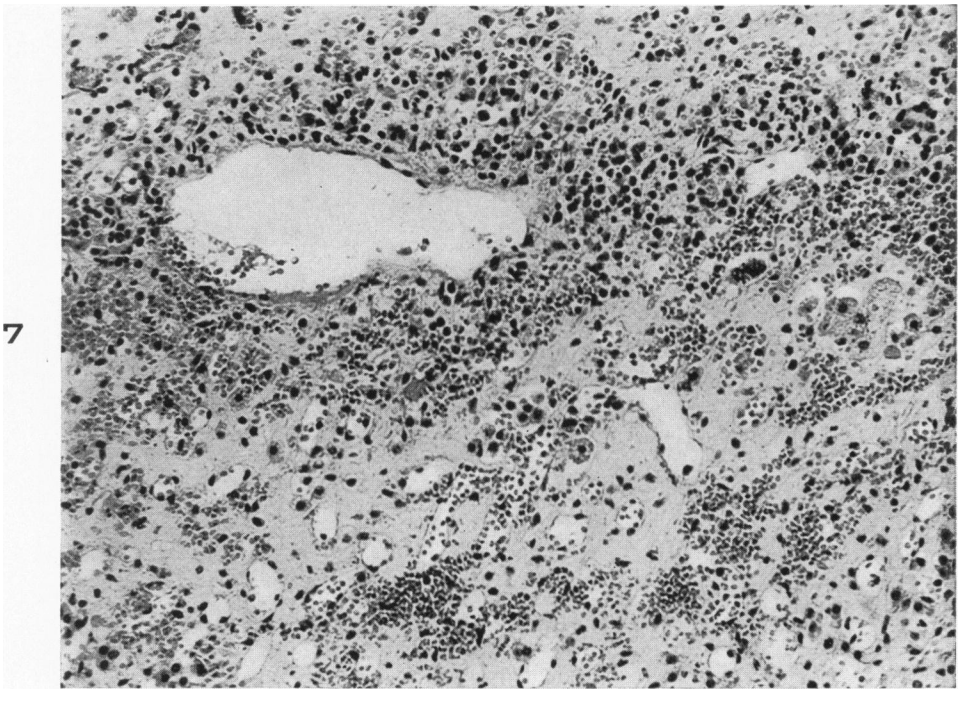


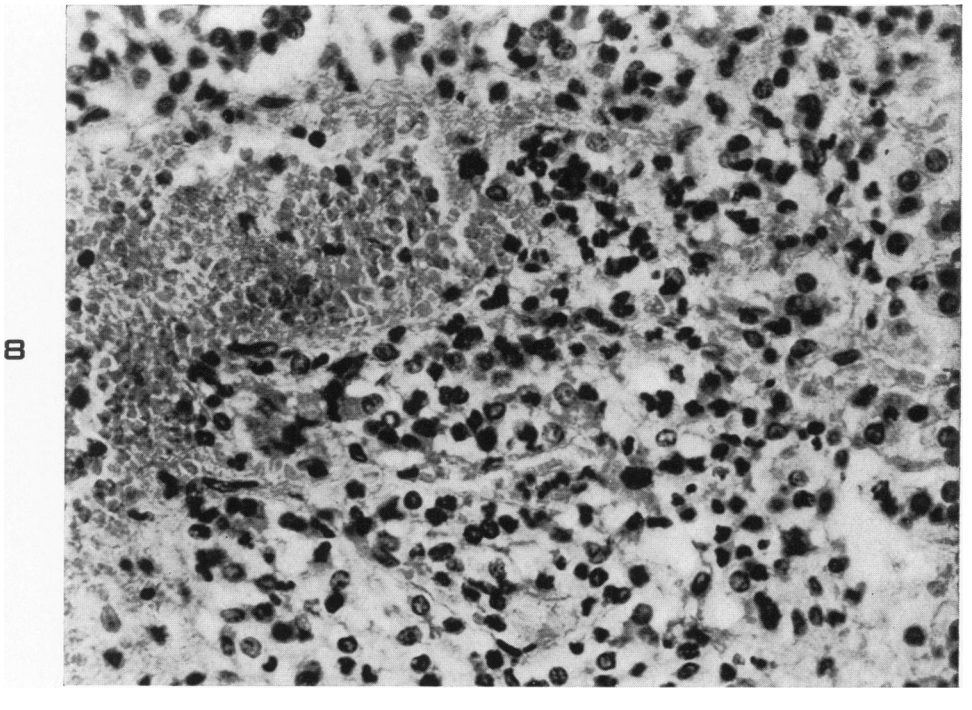
PLATE 126

FIG. 7. Adrenal medulla. Moderate perivascular infiltration of mononuclear cells with some neutrophils and eosinophils. Slight medullary hemorrhage. Hematoxylin and eosin stain. × 200.

FIG. 8. Adrenal medulla. Marked infiltration of neutrophils, eosinophils, and some mononuclear cells. Sparse filamentous fibrin thrombi in sinusoids. Very slight hemorrhage. Hematoxylin and eosin stain. × 530.



7



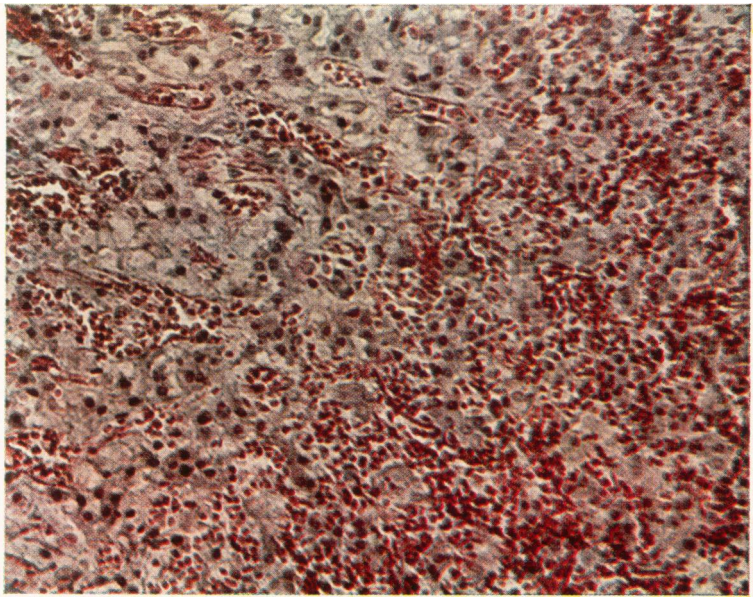
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PLATE 127

FIG. 9. Adrenal. Border between medulla and cortex. Rather marked hemorrhage. Most sinusoids are occluded by filamentous hyaline fibrin thrombi. Goldner's modification of Masson's trichrome stain. × 230.

FIG. 10. Adrenal as in Figure 9, showing filamentous hyaline thrombotic material extending along the occluding sinusoids. × 950.

9



10

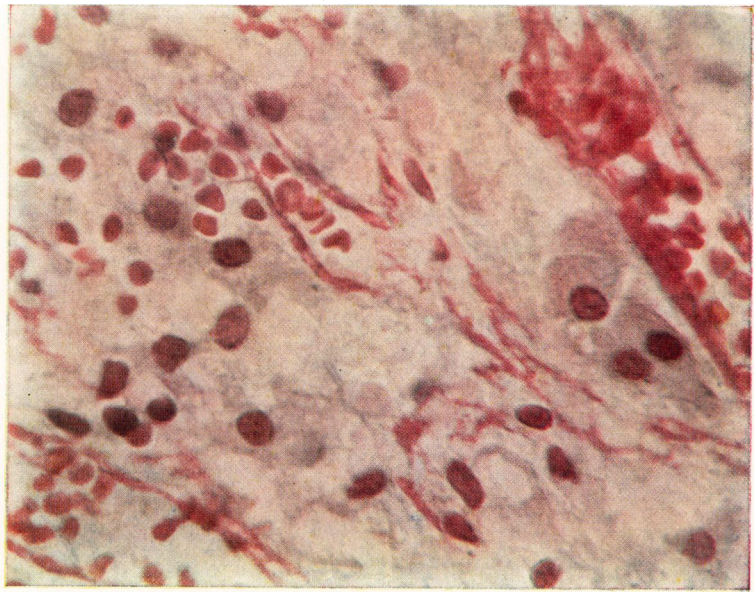
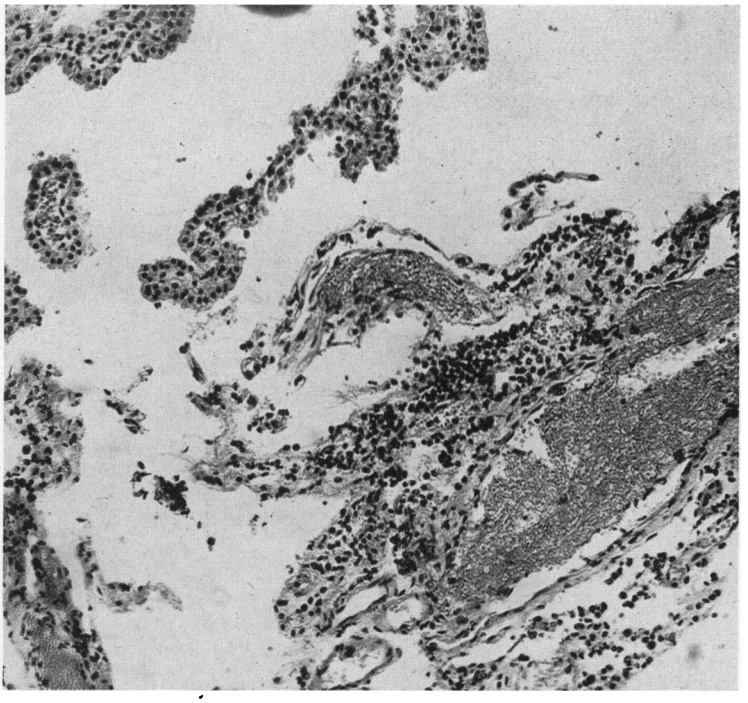


PLATE 128

FIG. 11. Choroid plexus. Inflammatory infiltration of interstitial tissue. Thrombosis of veins and sinusoids. Hematoxylin and eosin stain. × 115.

FIG. 12. Choroid plexus. Fibrin thrombus containing few cells. Goldner's modification of Masson's trichrome stain. × 115.

11



12

